JASP Student Developers

Introduction to QML for JASP

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2. QML files and QML coding
   - Coding challenges

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The JASP module

Folder structure

- Today, we'll focus on the inst folder
- More information about the general structure can be found [here](#).

```
.
├── <module_name>.Rproj
├── DESCRIPTION              # Describes the package and lists its dependencies
├── LICENSE
├── NAMESPACE                # Controls function importing
├── R                        # Where the package functions live
│   └── functions.R
│   └── more-functions.R
│   └── ...
├── README.md
├── renv.lock                # (Optional) Environment management...
├── _processedLockFile.lock # ...files, controlled by package renv
├── tests/                   # (Optional) Unit tests
│
│  # === So far, this is just a standard R package ===
│  # === Interaction with JASP starts below ===
│
└──inst
    ├── Description.qml      # Builds the ribbon menu
    ├── Upgrades.qml         # Optional
    ├──  qml                 # Folder containing one or more...
    │   └── analysis_1.qml   # ... module's menus
    │   └── ...
    ├── help                 # (Optional) Module's help files
    │   └── ...
    └── icons                # (Optional) Module's icons
        ├── <module_name>.svg
        └── ...
```

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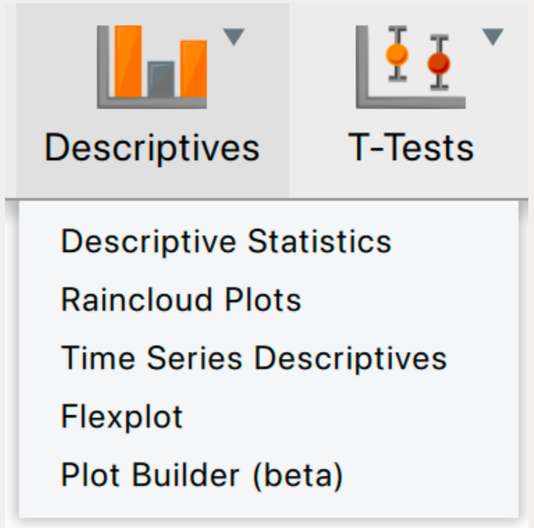
What's the function of the files?

inst/Description.qml



- Generates the ribbon menu
- Coordinates the other files to create the final analysis/ module
  - Assigns relevant R and QML files to the analyses
- Let's consider an example...

Objects in Description.qml

- Description
  - Provides the module name, description and icon
- Analysis
  - Name of the analysis, main R function and name of QML file for the analysis interface
- GroupTitle and Separator
  - Specify a title and visually split your analyses

Challenge (1)

The groundwork for your own analysis

- Fork the jaspModuleTemplate
- Create a new analysis
  - Don’t worry about the analysis QML files and R functions for now.
- Use a separator between the existing analyses and your new analysis
- Your analysis should be preceded by a group title

What’s the function of the files?

inst/qml/...

- Contains QML files for the analysis interfaces
- The Description.qml file links to these filenames
- You have a lot of freedom to tailor these files to your needs

How to write an analysis interface?

First steps

- Before you add QML components, your file needs imports...

```
import QtQuick
import QtQuick.Layouts
import JASP.Controls
import JASP.Widgets
```

- ... and a form that will contain the components

- It is also customary to copy over the copyright statement (you can find it in other files)

```
import QtQuick
import QtQuick.Layouts
import JASP.Controls
import JASP.Widgets

Form
{

}
```

How to write an analysis interface?

QML components

- Three classes
    1. General input (e.g., CheckBox)
    2. Variable assignment (e.g., AvailableVariablesList)
    3. Grouping components (e.g., Section)
- A thorough guide for QML components can be found here.

Challenge (2)

Add the QML file for your analysis

- Add a QML file for the analysis you added previously
- Link it to the analysis through the Description.qml file
- Add a checkbox to your interface